

L1.1b Transformations · §1.3

Outside or inside — what it moves, and which way

THE RULE

$$y = \underbrace{a}_{\text{R}} f\left(\underbrace{b}_{\text{D}} \left(x + \underbrace{c}_{\text{D}}\right)\right) + \underbrace{d}_{\text{R}}$$

“Anything outside the $f(x)$ affects **RANGE** normally. Anything inside the $f(x)$ affects **DOMAIN** backward.”

OUTSIDE | INSIDE

OUTSIDE the $f(x)$

⇒ moves **RANGE**, normally

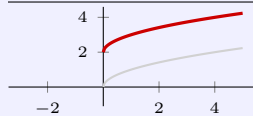
$f(x) + d$ slide d in the $+y$ direction

$a f(x)$, $|a| > 1$ stretch vertically

$a f(x)$, $0 < |a| < 1$ compress vertically

$-f(x)$ reflect over the x -axis

$|f(x)|$ flip whatever is below the x -axis up over it



$\sqrt{x} \rightarrow \sqrt{x} + 2$: R climbs to $[2, \infty)$, D never moves.

INSIDE the $f(x)$

⇒ moves **DOMAIN**, backward

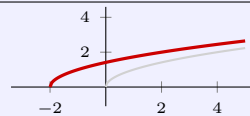
$f(x + c)$ slide c in the $-x$ direction

$f(bx)$, $|b| > 1$ **compress** horizontally

$f(bx)$, $0 < |b| < 1$ **stretch** horizontally

$f(-x)$ reflect over the y -axis

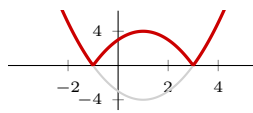
$f(|x|)$ chop off the negative x -values and reflect the positive values back over the y -axis



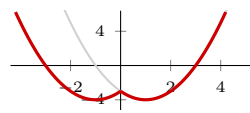
$\sqrt{x} \rightarrow \sqrt{x+2}$: D slides to $[-2, \infty)$ — the $+2$ went the *other* way.

THE ABSOLUTE VALUE PAIR — same bars, opposite sides

$f(x) = x^2 - 2x - 3$ — Loh: $m = 1$, $d = 2$, roots 1 ± 2 , so zeros at -1 and 3 and vertex $(1, -4)$.



$|f(x)|$ — **outside**. $R = \{y \mid y \geq 0\} = [0, \infty)$



$f(|x|)$ — **inside**. $R = \{y \mid y \geq -4\} = [-4, \infty)$

“outside bars bend the outputs up; inside bars chop and mirror the inputs”

FIVE COMBOS — A is f 's domain, B is g 's

$$f \pm g, \quad fg : \quad A \cap B \qquad \frac{f}{g} : \quad \{x \in A \cap B \mid g(x) \neq 0\}$$

“Intersection cuts out just the overlap of the two domain intervals; union gives you any place that either of them goes.”

COMPOSITION — diffnifod

$$\text{Domain}(f(g(x))) = D_{If} \cap I_{f \rightarrow OD}$$

$$x \rightarrow \boxed{g} \rightarrow g(x) \rightarrow \boxed{f} \rightarrow f(g(x))$$

$D_{If} \quad I_{f \rightarrow OD}$

D_{If} — the **inner** function has to be defined.

$I_{f \rightarrow OD}$ — the outer has to be defined too, but its input is $g(x)$:

push the outer domain **backward** through the inner.

“Domain of Inner Function, intersect Inner Function to Outer Domain — diffnifod”

Roles, not letters. In $g(f(x))$ the inner function is f , so D_{If} means f 's domain.

WATCH OUT

✗ ~~$f(x+2)$ shifts the graph 2 to the right~~ → **inside, so backward — slide 2 in the $-x$ direction**

✗ ~~$f(3x)$ stretches it horizontally by 3~~ → **inside, so backward — $|b| > 1$ compresses by 3**

✗ ~~$|f(x)|$ and $f(|x|)$ do the same thing~~ → **outside bars change outputs, inside bars change inputs**

✗ ~~the composite's domain is the outer function's domain~~ → **intersect both halves — D_{If} and $I_{f \rightarrow OD}$**